ξPLN: A Correct and Complete Foundation for Probabilistic Logic Networks (DRAFT)

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Abstract

Probabilistic Logic Networks (PLN) couples logical structure with uncertain inference. In practice, however, many PLN presentations mix distinct layers: (i) a probabilistic semantic model, (ii) an evidence/weight plumbing model, and (iii) operational truth-value views (strength, confidence, interval bounds). This leads to "semantic drift": algebraic operators are treated as if they were the logic itself, and link-level inference attempts to propagate correlation-sensitive quantities without carrying the information required to do so correctly.

We present ξPLN, a complete PLN foundation designed to be (a) mathematically correct and (b) extensible to tractable sublayers. The core move is to make the complete layer a *posterior-state calculus*: the proof object is a revisable world-model posterior state, and all link/event truth values are obtained by querying that state. We give a reference complete instance as a Dirichlet posterior over complete worlds (`JointEvidence`), prove that evidence extraction commutes with revision, and formalize a no-go theorem showing that complete link evidence cannot, in general, be computed from local per-link evidence alone. We then explain how Bayesian-network-style sublayers arise as restricted but tractable world-model classes, keeping PLN's operational rules as compiled tactics with explicit assumptions.

1 Motivation and Context

PLN, as described by Goertzel, Iklé et al. [7], represents uncertain judgments using truth values such as *strength* (a probability-like estimate) and *confidence* (a reliability-like quantity). Modern *nuPLN* work (Nil Geisweiller, draft and v1) makes the key step explicit: a complete grounding requires a global probabilistic semantics—a world model with a joint distribution over possible worlds and a Bayesian update story. We treat νPLN as the probabilistic reference layer and extend it in ξPLN by (i) making posterior states the proof objects, (ii) separating

evidence algebra from probability views, and (iii) connecting the complete layer to tractable Bayesian-network sublayers and Heyting-valued foundations [2].

Our experience formalizing PLN in Lean is that correctness failures almost always come from mixing layers:

- treating operational truth-value operators (interval bounds, confidence heuristics) as if they were the logic; or
- trying to compute complete posteriors for derived links from local link truth values without carrying the correlation information needed to do so.

ξPLN is a response: it isolates a *complete* core calculus and makes all fast PLN rules live as derived/compiled approximations relative to declared model classes. For the now-closed strength/confidence side of this story, the public capstone is `Mettapedia/PLN/Core/PLNCanonicalAPI.lean` under `confidenceCharacterizationEndpointProfil` the expository write-up is `papers/pln-credal-degrees-of-freedom.tex`. We keep this as a pointer only, not a second derivation.

2 Three Semantic Layers (and Why They Must Not Be Conflated)

We distinguish three layers.

2.1 Layer 1: World-model posterior states (complete semantics)

The complete layer carries a posterior state in some explicit class of world models. Conceptually: *a distribution or structured distribution* that can answer any query by conditioning/marginalization (or by exact inference within the chosen model class).

In the Lean prototype, this layer is represented by `JointEvidence`:

$$\texttt{JointEvidence}(n) := \texttt{Fin}(2^n) \to \mathbb{R}^{\infty}_{\geq 0}$$

interpreted as Dirichlet pseudo-counts over the 2^n complete worlds for n propositional atoms. Revision is pointwise addition of pseudo-counts.

2.2 Layer 2: Evidence algebra (quantale / Heyting structure)

PLN uses an evidence carrier `Evidence` as a compositional algebra of observations. In our formalization, `Evidence` is the 2D count object (n^+, n^-) with:

- parallel aggregation $\oplus$ ("revision"): add counts componentwise;
- order (information ordering): coordinatewise $\leq$;

- rich lattice structure (complete lattice; Heyting operators exist).

This layer is *not* itself the complete probabilistic semantics; it is an algebraic semantics for evidence aggregation and logical structure. It is particularly important for FO/HO extensions, where truth values are naturally Heyting-valued (cf. the SatisfyingSet construction in the codebase).

2.3 Layer 3: Operational views (strength, weight, confidence, bounds)

Strength/weight/confidence/intervals are *views* extracted from evidence. In particular, Walley-style predictive IDM intervals and Bayesian Beta-posterior credible intervals are distinct semantics and must remain explicit selector choices rather than being conflated into one operator [10, 11, 1]. They are indispensable operationally, but they should not be used as if they were complete semantic state. ξPLN adopts the mantra:

> Probability is what you query; evidence is what you carry.

3 ξPLN Core: World-model Revision + Query

The complete core calculus is intentionally simple: it is the calculus of revising posterior states plus a query interface.

3.1 World-model interface

In Lean, we isolate a minimal interface `WorldModel` (posterior state + query projections):

```
inductive PLNQuery (Atom : Type*) where
  | prop : Atom -> PLNQuery Atom
  | link : Atom -> Atom -> PLNQuery Atom
  | linkCond : List Atom -> Atom -> PLNQuery Atom

class WorldModel (State : Type*) (Query : Type*) [EvidenceType State]
    where
  evidence : State -> Query -> Evidence
  evidence_add : forall W1 W2 q, evidence (W1 + W2) q = evidence W1 q +
       evidence W2 q
```

Here `EvidenceType` means: revision is an additive commutative monoid on `State`. Standard PLN event/link queries are represented by `PLNQuery Atom`; all truth-value quantities are derived by mapping extracted `Evidence` to strengths/weights. See `Mettapedia/Logic/PLNWorldModel.lean`.

3.2 Reference complete instance: Dirichlet over worlds

For finite propositional scope (atoms `Fin n`), we instantiate `WorldModel` with the Dirichlet world-table `JointEvidence`:

- `evidence(E, prop(A))` sums world counts where an atom is `true` vs `false`;
- `evidence(E, link(A,B))` sums world counts where `A` is true and `B` true vs false.
- `evidence(E, linkCond([A,B],C))` sums world counts where both `A,B` hold and `C` holds vs fails.

Crucially, extraction commutes with revision:

> revising joint evidence and then extracting a query is equal to extracting and revising at the query-evidence level.

Formally, the Lean prototype proves:

$$\text{evidence}(E_1 + E_2, q) = \text{evidence}(E_1, q) \oplus \text{evidence}(E_2, q)$$

for all queries q. See `Mettapedia/Logic/PLNJointEvidence.lean`.

3.3 Probability views

From `Evidence` we obtain posterior-mean probabilities (improper-prior strength):

$$P(A) = \frac{\#(A)}{\#(\top)}, \qquad P(B \mid A) = \frac{\#(A \wedge B)}{\#(A)}$$

where $\#(\cdot)$ is a world-count sum in `JointEvidence`. The Lean file `Mettapedia/Logic/PLNJointEvidenceProbability.lean` proves the exact ratio forms for these views.

4 A Sequent-Calculus View: World Models and Links

We present ξPLN as a sequent-style system with two contexts:

- an *evidence context* Γ, a finite multiset of posterior fragments (elements of a world-model state type `State`); and
- a *side-condition context* Σ containing structural assumptions about the chosen world-model class (e.g. a BN DAG, positivity, d-separation facts).

4.1 Judgments

We use two judgment forms.

1. **World-model judgment:** $\Sigma; \Gamma \vdash_{\text{wm}} W$ meaning "from the evidence pieces in Γ, we can construct the revised posterior state W".

2. **Query judgment:** $\Sigma; \Gamma \vdash q \Downarrow e$ meaning "querying the revised posterior state derived from Γ yields evidence e for query q".

In the Lean prototype, the world-model judgment is deterministic: if `State` is an additive commutative monoid, then the constructed posterior is simply the revision sum $W := \sum \Gamma$. The query judgment is then obtained by extraction via `WorldModel`:

$$e := \text{evidence}(W, q).$$

4.2 Core rules

Let `State` be an `EvidenceType` (revision `+` with unit `0`), and let `WorldModel` provide evidence `: State → Query → Evidence`. The complete core calculus is:

$$\frac{}{\Sigma; \varnothing \vdash_{\text{wm}} 0} \text{ (WM-Unit)} \qquad \frac{}{\Sigma; \{W\} \vdash_{\text{wm}} W} \text{ (WM-Ev)} \qquad \frac{\Sigma; \Gamma \vdash_{\text{wm}} W \quad \Sigma; \Delta \vdash_{\text{wm}} W'}{\Sigma; \Gamma \uplus \Delta \vdash_{\text{wm}} (W + W')} \text{ (WM-Rev)}$$

$$\frac{\Sigma; \Gamma \vdash_{\text{wm}} W}{\Sigma; \Gamma \vdash q \Downarrow \text{evidence}(W, q)} \text{ (Q-Extract)}$$

The key algebraic law is commutation of extraction with revision:

$$\text{evidence}(W_1 + W_2, q) = \text{evidence}(W_1, q) \oplus \text{evidence}(W_2, q),$$

which makes the following *query-revision* rule admissible:

$$\frac{\Sigma; \Gamma \vdash q \Downarrow e \quad \Sigma; \Delta \vdash q \Downarrow e'}{\Sigma; \Gamma \uplus \Delta \vdash q \Downarrow (e \oplus e')} \text{ (Q-Rev)}$$

Links are queries. In this view, a PLN "link" $A \Rightarrow B$ is not an object-level connective; it is a query `link(A,B)` against the revised world model. The calculus does not propagate links directly; it revises world models and answers link queries.

4.3 Derived link calculus: classic PLN rules as admissible rewrites

Classic PLN rules (deduction/abduction/induction) become *compiled tactics* or *admissible query rewrites* relative to a world-model class with explicit side conditions Σ. Intuitively, they are ways to answer some link queries using other link/event queries, without pretending to compute complete link evidence in full generality.

Example: deduction strength admissibility (BN case). Let $\mathsf{strength}(W, q)$ denote the posterior-mean view extracted from evidence:

$$\mathsf{strength}(W, q) := \text{toStrength}(\text{evidence}(W, q)).$$

For a BN world-model class, if Σ entails the screening-off equalities needed by the general PLN deduction theorem (conditional independence of C from A given B and given $\neg B$, plus positivity), then one may rewrite a hard query `linkCond([A,B],C)` into smaller ones:

$$\mathsf{strength}(W, \texttt{linkCond([A,B],C)}) = \text{plnDeductionStrength}\Big(\mathsf{strength}(W, \texttt{link}(A,B)), \mathsf{strength}(W, \texttt{link}(B,C)), \mathsf{str}$$

Lean status. `Mettapedia/Logic/PLNBayesNetFastRules.lean` proves this admissibility in the simplest nontrivial BN instance, the chain $A \to B \to C$: the theorem `chainBN_plnDeductionStrength_exact` shows the PLN deduction strength formula computes the exact $P(C \mid A)$ in the chain BN under explicit positivity side conditions. The general BN-facing rewrite layer uses the query-level screening-off form

$$\texttt{linkCond([A,B],C)} = \texttt{link(B,C)}$$

under explicit semantic obligations (see `Mettapedia/PLN/Bridges/ProbabilityTheory/BayesNet/PLNBNCompilation.lean`).

Admissible link-rule schema (strength view). Let $\mathsf{strength}(e)$ denote the posterior-mean view extracted from evidence e. Then, for any world-model class where Σ entails the screening-off conditions, the following rule is admissible:

$$\frac{\Sigma \vdash \mathsf{ScreenOff}(A,B,C) \quad \Sigma;\Gamma \vdash \texttt{link}(A,B) \Downarrow e_{AB} \quad \Sigma;\Gamma \vdash \texttt{link}(B,C) \Downarrow e_{BC} \quad \Sigma;\Gamma \vdash \texttt{prop}(B) \Downarrow e_B \quad \Sigma;\Gamma \vdash \texttt{prop}(}{\Sigma;\Gamma \vdash \texttt{linkCond([A,B],C)} \Downarrow e_{AC}}$$

with the side-condition that

$$\mathsf{strength}(e_{AC}) = \text{plnDeductionStrength}(\mathsf{strength}(e_{AB}), \mathsf{strength}(e_{BC}), \mathsf{strength}(e_B), \mathsf{strength}(e_C)).$$

This is a *strength-level* rewrite: the evidence e_{AC} itself is still obtained by querying the world model, unless an explicit evidence-flow law is available for the given model class.

5 A No-Go Theorem: "Complete" Link Inference Cannot Be Local

One might hope for a sequent-calculus-like system that takes local per-link evidence (`Evidence` for A, B, C, $A \Rightarrow B$, $B \Rightarrow C$) and computes complete evidence for $A \Rightarrow C$. ξPLN asserts that this is impossible in general without extra assumptions: correlations live in the joint state.

Theorem 1 (No local complete deduction rule)**.** *There is no function*

$$f : \texttt{Evidence}^5 \to \texttt{Evidence}$$

that, for all joint evidence states E, computes the exact link evidence for $A \Rightarrow C$ from only the local premises $\texttt{Evidence}(A), \texttt{Evidence}(B), \texttt{Evidence}(C), \texttt{Evidence}(A \Rightarrow B), \texttt{Evidence}(B \Rightarrow C)$.

The Lean proof constructs two different joint evidence states on three atoms that agree on all these premises but disagree on the conclusion `linkEvidence(A,C)`. See `Mettapedia/Logic/PLNJointEvidenceNoGo.lean`.

Interpretation. This theorem is the formal core of "you must carry correlations". It does not make fast PLN useless; it tells us exactly what fast PLN *cannot* claim: completeness for arbitrary world models without structural assumptions.

6 Case Study: Markov Carriers and the Screening No-Go

We instantiate the ξPLN carrier framework for Markov transition evidence, revealing a fundamental incompatibility between PLN's additive deduction and Markov transition composition.

6.1 Single-Step Semantics: Success

The `markovDirichletQueryCarrier` (`Mettapedia/PLN/Bridges/ProbabilityTheory/PLNXiCarrierScreening.lean:483`) provides query-indexed projection from row evidence to binary "target vs rest" evidence. Single-step queries work correctly: the atom `(markov-transition i j)` evaluates to row-i evidence projected at coordinate j.

6.2 Multi-Step Screening: Failure

The screening condition for PLN deduction is:

$$\text{proj}(A{\to}C, \text{rowA} + \text{rowB}) = \text{proj}(A{\to}C, \text{rowA})$$

This requires rowB to contribute zero at coordinate C—false in general. The additive carrier is *structurally wrong* for Markov composition.

6.3 Honest Alternative: Predictive Chaining

We formalize the correct algebra via `markovWMPosteriorChain` (`MarkovPredictiveChaining.lean:32`):

1. Singleton `MassSemantics` state (not additive)
2. Query mass = Chapman-Kolmogorov product over paths
3. Path atoms compile to predictive chain, not additive pooling

The `markov-transition*` atom family (`MarkovPathXi.lean:43`) provides the OSLF API with concrete binary examples proving end-to-end correctness (`bit001_queryStrength_transitionMultiset_eq_of_summary`).

6.4 Lesson for PLN Design

When evidence doesn't compose additively, don't force it through additive carriers. Design specialized interfaces with honest semantics. The PeTTaChainer implementation validates this: its `MpFormula` uses strength multiplication $s_{AB} \cdot s_{BC}$, not addition—consistent with our mathematical finding.

7 Tractable Sublayers: Bayesian Networks as Restricted World Models

The complete world-table `JointEvidence` is exponential in n. To scale, we restrict the world-model class while preserving correctness *relative to that class*.

The next natural step is a Bayesian-network-style world model:

- a DAG structure specifying factorization;
- local conditional tables with Dirichlet evidence (counts) per CPT row;
- revision = add local Dirichlet counts (conjugate update);
- query = exact inference by variable elimination / junction tree, with complexity controlled by treewidth.

Lean status. We have implemented the *evidence plumbing* part of this sublayer:

- a Boolean BN CPT query type `CPTQuery` (node + parent configuration),
- a CPT posterior state `CPTState` storing `Evidence` per CPT entry, and
- an additive projection from `JointEvidence` to CPT evidence by marginalization (summing compatible worlds), which commutes with revision.

See `Mettapedia/PLN/WorldModel/BayesNet/PLNBayesNetWorldModel.lean`.

We have also begun the "fast rule exactness" bridge in the simplest nontrivial case: `Mettapedia/Logic/PLNBayesNetFastRules.lean` sets up the chain BN $A \to B \to C$, proves the required sink-factorization and normalization lemmas, derives the screening-off hypotheses, and applies `PLNDerivation.pln_deduction_from_total_probability` to obtain an *exactness theorem*: in the chain BN, the standard PLN deduction strength formula computes the correct conditional probability $P(C \mid A)$ (under the explicit positivity side-conditions).

Exact BN query answering (variable elimination / junction tree) is the next step: it will provide tractable evaluation of probabilities for larger query languages, while preserving correctness relative to the declared BN model class.

PLN spirit preserved. Classic PLN link rules (deduction/abduction/induction) become:

- compiled tactics that propose BN queries, or
- conditionally-sound lemmas under explicit assumptions (e.g. conditional independence),

rather than pretending to be globally complete link-level operators.

8 Factor Graphs: Semantic Factorization vs Operational Control

PLN has long used factor-graph and message-passing intuition for efficient inference. The recent MORK/MM2+ notes make this explicit by proposing a *Quantale-Annotated PLN Factor Graph* encoding in Atomspace (variables = formulas with truth values; factors = rule instances/potentials; messages combined with $\oplus$ and $\otimes$) [6, 5]. This is the right operational substrate, but ξPLN separates two distinct roles:

8.1 (A) Semantic factor graphs (world-model factorization)

Here factors *are* the model: the world-model posterior state is represented by a factorization of a joint distribution (BN or MRF). In this case:

- factors are genuine conditional / clique potentials;
- variable elimination / junction tree gives exact query answers (treewidth controls complexity);
- belief propagation is exact on trees and approximate on loopy graphs.

Thus a factor graph is a *world-model representation*, and correctness is relative to the declared model class.

Positive example (semantic). A BN with edges $A \to B \to C$ yields factors $P(A)$, $P(B \mid A)$, $P(C \mid B)$. If the model class is fixed to this DAG, variable elimination returns the exact $P(C \mid A)$ and the PLN deduction strength formula is admissible as a query rewrite.

8.2 (B) Operational factor graphs (inference control)

Here factors are *rule schemas* (deduction, induction, revision, etc.) whose local "potentials" are truth-value update formulas. This aligns with the PLN chainer and MORK PathMap indexing [6, 4]. However, unless each factor can be justified as a true conditional or likelihood factor in a world model, this structure should be viewed as *inference control*: it proposes which queries to ask and which rewrites to attempt, but it does not itself constitute the semantics.

Negative example (operational). A graph whose factors are "deduction" or "abduction" update formulas on STVs is useful for search, but it is not automatically a probabilistic model: unless each factor corresponds to a conditional/likelihood in some WM class, message passing is a heuristic scheduler rather than an exact semantics.

8.3 Alignment: when the two coincide

The two notions align when rule factors are *compiled* from a world-model class: each rule corresponds to a conditional or likelihood factor, and message passing implements the same queries that the world-model semantics would answer. This is exactly the intended "fast PLN rules as compiled tactics" story: the factor graph becomes a *query plan* whose correctness is discharged by explicit side conditions (Σ) such as d-separation or Markov properties.

Evidence accounting and control. Geodesic inference control suggests a decentralized evidence ledger to prevent double counting during message passing and chaining [3]. This is naturally compatible with the `WorldModel` interface: evidence is revised at the world-model layer, while factor-graph message passing is used to schedule or approximate queries, with provenance tracking guarding independence assumptions.

9 Knuth–Skilling and Heyting Foundations

Knuth–Skilling foundations of inference [8] provide axioms for valuation schemes on logical lattices, extending Cox-style plausibility calculi. In ξPLN, this plays two roles:

1. *Probability as a valuation/view.* At the world-model layer, probabilities arise as valuations on the Boolean algebra of subsets of worlds. This is the classical Kolmogorov story [9] instantiated for finite spaces, and it underwrites the ratio theorems proved in `PLNJointEvidenceProbability`.

2. *Evidence as Heyting-valued semantics.* At the evidence/algebraic layer, `Evidence` forms a rich (non-Boolean) lattice. K&S-style valuation rules persist but Boolean equalities can weaken to inequalities; see `Mettapedia/Logic/EvidenceIntuitionisticProbability.lean`.

We also bridge K&S additive regraduations to valuation-algebra factor graphs, so VE can be run end-to-end on KS-valued factors without assuming probability normalization; see `Mettapedia/ProbabilityTheory/KnuthSkilling/Bridges/ValuationAlgebra.lean` and `Mettapedia/ProbabilityTheory/BayesianNetworks/KSFactorGraph.lean`.

An important cautionary fact—the "totality gate"—is already formalized: because `Evidence` has incomparable elements, it admits no faithful order-embedding into the reals. Thus one should not expect a single real-valued map to capture all evidence information. See `Mettapedia/Logic/PLN_KS_Bridge.lean`.

10 Roadmap and Deliverables

The core Lean artifacts supporting ξPLN are:

- `Mettapedia/Logic/PLNWorldModel.lean`: the world-model interface and derived views;
- `Mettapedia/Logic/PLNJointEvidence.lean`: the reference complete instance and revision-commutes-with-extraction;
- `Mettapedia/PLN/WorldModel/BayesNet/PLNBayesNetWorldModel.lean`: BN-style CPT evidence states and the additive projection from `JointEvidence`;
- `Mettapedia/Logic/PLNBayesNetFastRules.lean`: chain BN proof that the PLN deduction strength formula is exact in $A \to B \to C$;
- `Mettapedia/Logic/PLNJointEvidenceProbability.lean`: posterior-mean probability views (ratio theorems);
- `Mettapedia/PLN/Bridges/ProbabilityTheory/BayesNet/PLNBNCompilation.lean`: BN query compilation, `linkCond` queries, and Σ-guarded screening-off rewrites;
- `Mettapedia/Logic/PLNWorldModelITV.lean`: explicit ITV semantics layer (Bayes-normal, Bayes-exact, Walley-IDM);
- `Mettapedia/Logic/PLNWorldModelITVHypercube.lean`: interval-selector transport and typed ITV judgment plumbing;
- `Mettapedia/PLN/Bridges/Languages/PLNWMOSLFBridgeITV.lean`: typed OSLF ITV bridge and selector-specialized coherence theorems;
- `Mettapedia/PLN/RuleFamilies/FirstOrder/PLNXiDerivedBNRules.lean`: typed BN constructor lifts (unit/constant/dependent/native/sort-tagged);
- `Mettapedia/PLN/Core/PLNCanonicalAPI.lean`: canonical one-call end-to-end endpoints and final bundle APIs;
- `Mettapedia/OSLF/Framework/QuantaleCoherence.lean`: reusable quantale transport/coherence bundle;
- `Mettapedia/ProbabilityTheory/BayesianNetworks/VariableElimination.lean`: exact VE engine;
- `Mettapedia/ProbabilityTheory/BayesianNetworks/ValuationAlgebra.lean`: valuation-algebra semantics + VE correctness spine;
- `Mettapedia/ProbabilityTheory/BayesianNetworks/VEBridge.lean`: joint-measure ↔ VE probability bridge;
- `Mettapedia/ProbabilityTheory/KnuthSkilling/Bridges/ValuationAlgebra.lean`: KS → valuation-algebra bridge;

- `Mettapedia/Logic/PLNJointEvidenceNoGo.lean`: the local-completeness no-go theorem.

Recent ITV/typed-bridge deliverables now in the Lean stack:

1. explicit separation of Walley predictive intervals [11] from Bayesian credible intervals at the semantics layer;
2. typed WM-to-OSLF bridge theorems that expose coordinate-wise threshold transport (lower/upper/credibility/width/strength);
3. one-call composed endpoints that chain selector transport, rewrite/query transport, and quantale coherence for chain/fork/collider BN constructors.
4. intensional inheritance now has a typed WM/query layer and Solomonoff-linked semantic transport: `Mettapedia/Logic/PLNIntensionalWorldModel.lean`, `Mettapedia/Logic/IntensionalInheritanceSolomonoffBridge.lean`, and linked one-call final-bundle endpoints in `Mettapedia/PLN/Core/PLNCanonicalAPI.lean`.

Next implementation milestones:

1. implement exact BN query answering (variable elimination first) for prop/link queries over the BN/factor-graph world-model layer;
2. generalize chain exactness to arbitrary BNs via d-separation/Markov properties so screening-off obligations are discharged structurally;
3. add a compilation layer: PLN rule patterns $\Rightarrow$ BN query plans with explicit Σ side-conditions.
4. extend query language beyond atoms to formulas/events, enabling FO/HO semantics via SatisfyingSets while keeping probability views as world-model queries.
5. stabilize sort-tagged native-path coordinate-specialized final-bundle aliases (chain/fork/collider for lower/upper/credibility/width/strength);
6. close remaining proof debt in confidence-space soundness layers so all downstream one-call APIs rest on fully discharged theorem dependencies.
7. **Chapter 9 future work (operational closure)**: complete the operational large-scale inference layer once the Lean-side MeTTa runtime stack is stable enough for executable chainer/runtime integration; until then, Ch. 9 remains theorem-level and counterexample-first on the Lean side.

11 Conclusion

ξPLN formalizes the slogan "pass distributions, not just truth values" in a way that is precise enough to support soundness/completeness claims and flexible enough to support tractable sublayers. The complete layer is not a competing "second PLN"; it is the reference semantics that operational PLN rules approximate, under explicit assumptions, within declared model classes.

A PLN Book Coverage Map

This appendix maps the core units of the PLN book ([7]) to the current Lean formalization and indicates how each unit is represented. Coverage labels are:

- **Formalized**: theorem-level core is present in the main stack.
- **Partial**: substantial core is formalized, but some book-era variants remain open.
- **POC**: exploratory/prototype formalization, not yet integrated as a complete layer.

Book unit	**Core content**	**Lean coverage (where/how)**	**Status**
Ch. 1	Introduction	Conceptual framing is encoded as an explicit semantic stack split: world-model posterior semantics vs evidence algebra vs operational truth-value views (this paper; `Mettapedia/PLN/Core/PLNCore.lean`, `Mettapedia/Logic/SemanticsDecisionTree.lean`).	Formalized
Ch. 2	Knowledge representation	Query-level KR via `PLNQuery`, `WorldModel`, `WorldModelSigma`, and WM rewrite judgments: `Mettapedia/Logic/PLNWorldModel.lean`, `Mettapedia/Logic/PLNWorldModelCalculus.lean`, `Mettapedia/PLN/Bridges/ProbabilityTheory/BayesNet/PLNBNCompilation.lean`.	Formalized

Ch. 3	Experiential semantics	Evidence-as-counts carrier and revision algebra: `Mettapedia/Logic/EvidenceClass.lean`, `Mettapedia/Logic/EvidenceQuantale.lean`; complete posterior state semantics via `Mettapedia/Logic/PLNJointEvidence.lean`.	Formalized
Ch. 4	Indefinite truth values	ITV core plus explicit semantics split: `Mettapedia/Logic/PLNIndefiniteTruth.lean` (including Walley IDM predictive interval constructors), `Mettapedia/Logic/PLNWorldModelITV.lean`, `Mettapedia/Logic/PLNWorldModelITVHypercube.lean`.	Formalized
Ch. 5	FO extensional inference (rules/strength formulas)	Core deduction/induction/abduction formulas and derived rules: `Mettapedia/Logic/PLNDeduction.lean`, `Mettapedia/PLN/RuleFamilies/FirstOrder/PLNDerivation.lean`, `Mettapedia/Logic/PLNInferenceRules.lean`, `Mettapedia/Logic/PeTTaLibPLNTruthFunctions.lean`, `Mettapedia/Logic/WMPLNJustifiedTruthFunctions.lean`, `Mettapedia/PLN/Comparisons/PeTTa/PeTTaLibPLNFormalAnalysis.lean`; BN exactness anchors: `chainBN_plnDeductionStrength_exact`, `forkBN_plnDeductionStrength_exact` in `Mettapedia/Logic/PLNBayesNetFastRules.lean`.	Formalized

Ch. 6	FO extensional inference with ITV	ITV semantics are transported through WM rewrites, typed OSLF bridge, and one-call API bundles: `Mettapedia/PLN/Bridges/Languages/PLNWMOSLFBridgeITV.lean`, `Mettapedia/PLN/Core/PLNCanonicalAPI.lean`.	Formalized
Ch. 7	Distributional truth values	Beta/Dirichlet distributional layer: `Mettapedia/Logic/PLNDistributional.lean`, `Mettapedia/PLN/Bridges/ProbabilityTheory/EvidenceBeta.lean`, `Mettapedia/PLN/Bridges/ProbabilityTheory/EvidenceDirichlet.lean`; Kyburg higher-order flattening and decision-equivalence bridge is theorem-level and exposed at `Mettapedia/Logic/HigherOrder/PLNKyburgReduction.lean` with canonical aliases in `Mettapedia/PLN/Core/PLNCanonicalAPI.lean`.	Formalized (core + Kyburg bridge)
Ch. 8	Error magnification	Confidence-level WM transport and OSLF-threshold grounding: `Mettapedia/PLN/Bridges/Languages/PLNErrorMagnificationGrounding.lean`, canonical one-call endpoints in `Mettapedia/PLN/Core/PLNCanonicalAPI.lean` (including `wm_rewrite_oslf_threshold_acceptance_of_interval`); diagnostic/characterization layers remain in `Mettapedia/Logic/PLNBugAnalysis.lean`, `Mettapedia/Logic/Comparison/ErrorCharacterization.lean`.	Formalized (core)

Ch. 9	Large-scale inference strategies	Two tracks: (i) structural BN compilation/rewrite pipeline (`Mettapedia/PLN/Bridges/ProbabilityTheory/BayesNet/PLNBNCompilation.lean`, `Mettapedia/PLN/RuleFamilies/FirstOrder/PLNXiDerivedBNRules.lean`); (ii) premise-selection/inference-control theory (`Mettapedia/PLN/InferenceControl/PremiseSelection/SelectorSpec.lean`, `Mettapedia/PLN/InferenceControl/PremiseSelection/Optimality.lean`, `Mettapedia/PLN/InferenceControl/PremiseSelection/RankingStability.lean`, `Mettapedia/PLN/InferenceControl/PremiseSelection/Coverage.lean`). Counterexample-first no-go/corrected-positive index: `Mettapedia/PLN/InferenceControl/Counterexamples/PLNLargeScaleInferenceCounterexamples.lean`. Operational runtimes (execution-engine implementations and performance heuristics) are pending operational Lean-MeTTa runtime developments in this chapter map.	Partial (counterexample-first core formalized; operational pending Lean-MeTTa runtime)
Ch. 10	Higher-order extensional inference	HOI→FOI reduction formalized: `Mettapedia/Logic/HigherOrder/Basic.lean`, `Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`, `Mettapedia/Logic/HigherOrder.lean`.	Formalized (core)

Ch. 11	Crisp/fuzzy quantifiers with indefinite TVs	Quantifier semantics and weakness connection: `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers.lean`, `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/WeaknessConnection.lean`, `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/Soundness.lean`; fuzzy quantifier algebra and ITV-coordinate bridge: `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzyQuantifierSemantics.lean`, `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzyITVBridge.lean`; theorem pointers: `canary_rule_family_end_to_end_ext`, `canary_existentialConjunction_not_equivalent_itvPath`, `fuzzyIntervalHolds_strength_of_lower_upper`, `canary_itv_strength_interval_of_lower_upper`; crisp operators: `Mettapedia/Logic/PLNCrispEvidence.lean`, `Mettapedia/Logic/PLNNoisyOr.lean`; regression target: `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierRegression.lean`, `lake build Mettapedia.PLN.RuleFamilies.FirstOrder.Quantifiers.QuantifierRegression`.	Formalized (core + fuzzy/QFM + regressions)

Ch. 12	Intensional inference	Information-theoretic bridge and typed WM grounding: `Mettapedia/Logic/IntensionalInheritance.lean`, `Mettapedia/Logic/IntensionalInheritanceSolomonoffBridge.lean`, `Mettapedia/Logic/PLNIntensionalWorldModel.lean`, semantic and linked one-call endpoints in `Mettapedia/PLN/Core/PLNCanonicalAPI.lean`, plus executable canary/regression targets: `Mettapedia/Logic/PLNIntensionalCanary.lean`, `Mettapedia/Logic/PLNIntensionalRegression.lean`, `lake build Mettapedia.PLN.ConceptGeometry.AssocPat.PLNIntensionalRegression`.	Formalized (ASSOC core)

Ch. 13	Aspects of inference control	Formal inference-control components via selector specification, ranking transfer/stability, aggregation-role, and executable canary/regression targets: `Mettapedia/PLN/InferenceControl/PremiseSelection/SelectorSpec.lean`, `Mettapedia/PLN/InferenceControl/PremiseSelection/Optimality.lean`, `Mettapedia/PLN/InferenceControl/PremiseSelection/RankingStability.lean`, `Mettapedia/PLN/InferenceControl/PremiseSelection/Coverage.lean`, `Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlCore.lean`, `Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlCanary.lean`, `Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlRegression.lean`, `lake build Mettapedia.PLN.InferenceControl.PremiseSelection.PLNInferenceControlRegression`.	Formalized (core + canaries)

Ch. 14	Temporal and causal inference	Temporal operator layer and causal profile: `Mettapedia/PLN/RuleFamilies/Temporal/PLNTemporalCausalInference.lean`; event-calculus WM grounding: `Mettapedia/PLN/RuleFamilies/Temporal/PLNProbabilisticEventCalculus.lean`; temporal OSLF/GSLT and one-call event endpoints: `Mettapedia/PLN/Core/PLNCanonicalAPI.lean` (event bundle/final-bundle endpoints).	Formalized (core)
App. A	Comparison of PLN and NARS rules	Consolidated theorem bundle: `Mettapedia/PLN/Comparisons/NARS/PLNNARSRuleCorrespondence.lean`, with source bridges `Mettapedia/PLN/Comparisons/NARS/NARSEvidenceBridge.lean` and `Mettapedia/PLN/Comparisons/NARS/NARSPLNGaloisConnection.lean`.	Formalized

The table is intentionally conservative: rows marked *Partial* indicate that the core semantic/theorem spine is present, while some book-era variants (or full-scale operational integration) are still being extended.

Coverage delta (current). Relative to earlier drafts, Ch. 12 has moved from prototype-only status to *formalized ASSOC+PAT semantic coverage*: typed WM channel grounding, canonical `AssocPatSemanticModel` packaging, selector-specialized end-to-end endpoints, and a theorem-level no-go witness (`no_binary_mixed_policy_for_assocPat`) that rules out collapsing the concrete ASSOC+PAT mixed channel to a universal binary extensional+ASSOC law.

A.1 Chapter 7 Rule-to-Theorem Index

Chapter 7 content	**Lean theorem(s)**	**Module**
Context-aware strength as Beta posterior mean coordinates *Tags: [PLNBook-Ch7], [Beta], [distributional]*	`strengthWith_eq_beta_posterior_mean`, `evidence_encodes_beta_parameters`	`Mettapedia/ENNReal/HigherOrder/PLNKyburgReduction.lean`

Evidence aggregation as conjugate update *Tags: [PLNBook-Ch7], [conjugacy], [revision]*	`hplus_is_bayesian_update`, `evidence_aggregation_is_conjugate_update`	`Mettapedia/Logic/HigherOrderEvidenceUpdate PLNKyburgReduction.lean`, `Mettapedia/PLN/Bridges/ProbabilityTheory/EvidenceBeta.lean`
Kyburg flattening and expectation-preserving reduction *Tags: [PLNBook-Ch7], [Kyburg], [higher-order], [flattening]*	`kyburg_flattening`, `expectation_consistency`, `kyburg_no_advantage`, `kyburg_no_advantage_via_mean`	`Mettapedia/Logic/HigherOrder/PLNKyburgReduction.lean` and `Mettapedia/ProbabilityTheory/HigherOrderProbability/KyburgFlattening.lean`, `Mettapedia/ProbabilityTheory/HigherOrderProbability/GiryMonad.lean`
Monad associativity for flattening (typed kernel convenience) *Tags: [PLNBook-Ch7], [Kyburg], [Giry], [associativity]*	`flatten_associativity`, `flatten_associativity_kernel`, `flatten_associativity_kernel'`	`Mettapedia/ProbabilityTheory/HigherOrderProbability/GiryMonad.lean`, `Mettapedia/PLN/Core/PLNCanonicalAPI.lean`
De Finetti singleton bridge into Kyburg flattening *Tags: [PLNBook-Ch7], [DeFinetti], [mixture]*	`deFinetti_flatten_apply_singleton`, `evidenceAggregation_and_kyburgFlattening`	`Mettapedia/Logic/HigherOrderEvidenceUpdate PLNKyburgReduction.lean`
Concrete worked posterior-mean fixture (uniform prior, evidence 3:1) *Tags: [PLNBook-Ch7], [worked-example], [Beta]*	`uniformPrior_strength_three_quarters`	`Mettapedia/Logic/HigherOrder/PLNKyburgReduction.lean`

A.2 Chapter 8 Rule-to-Theorem Index

Chapter 8 content	Lean theorem(s)	Module

View transport under WM query equivalence (confidence / strengthWith / interpret) *Tags: [PLNBook-Ch8], [WM-queryEq], [view-transport]*	WMQueryEq.to_queryStrengthWith, WMQueryEq.to_queryStrengthWith_threshold, WMQueryEq.to_queryConfidence, WMQueryEq.to_queryConfidence_threshold, WMQueryEq.to_queryInterpret, WMQueryEqSigma.to_queryStrengthWith, WMQueryEqSigma.to_queryStrengthWith_threshold, WMQueryEqSigma.to_queryConfidence, WMQueryEqSigma.to_queryConfidence_threshold, WMQueryEqSigma.to_queryInterpret	Mettapedia/Logic/PLNWorldModelCalculus.lean
Weaker bridge notions (evidence preorder + view equivalence) *Tags: [PLNBook-Ch8], [bridge], [evidence-order], [viewEq]*	WMQueryEq.to_evidenceLE, WMEvidenceLE.antisymm_to_WMQueryEq, WMQueryEq.to_viewEq, WMQueryEqSigma.to_evidenceLE, WMEvidenceLESigma.antisymm_to_WMQueryEqSigma, WMQueryEqSigma.to_viewEq	Mettapedia/Logic/PLNWorldModelCalculus.lean
Confidence-threshold atom semantics and rewrite soundness *Tags: [PLNBook-Ch8], [threshold-semantics], [OSLF]*	thresholdAtomSemOfWMQConfidence, thresholdAtomSemOfWMQSigmaConfidence, wmRewriteRule_threshold_sound, wmRewriteRuleSigma_threshold, rewrite_then_queryEq_threshold_atom_confidence, rewrite_then_queryEq_threshold_atom_confidence_sigma	Mettapedia/PLN/BridgingLanguages/PLNErrorMagnificationGrounding.lean
Threshold-bridge boundary (disjunction/diamond) and total-order recovery *Tags: [PLNBook-Ch8], [boundary], [counterexample], [totality]*	threshold_or_counterexample, threshold_dia_fails, threshold_or_total, threshold_dia_total	Mettapedia/Logic/OSLFEvidenceSemantics.lean
Double-damping threshold-gap characterization *Tags: [PLNBook-Ch8], [double-damping], [gap]*	double_damping_strict_lt, threshold_gap_of_double_damping	Mettapedia/PLN/BridgingLanguages/PLNErrorMagnificationGrounding.lean
One-call WM rewrite → OSLF truth transport → threshold acceptance *Tags: [PLNBook-Ch8], [end-to-end], [canonical-API]*	thresholdAccepted, thresholdAccepted_of_finalBundle, wm_rewrite_oslf_threshold_canonical_API, thresholdAccepted_bool_sound, end_to_end_quantale_selector_query_threshold_finalBundle_of_interval,	Mettapedia/PLN/InferenceControl/SelectorRewriteThreshold/PLNSelectorRewriteThresholdExamples.lean

WM consequence-rule layer and Kripke deontic lifts (strength inequalities) *Tags: [PLNBook-Ch8], [WMConsequenceRule], [Kripke], [modal-lift]*	`WMStrengthLE`, `WMConsequenceRule`, `WMConsequenceRuleOn`, `WMConsequenceRuleOn.ofGlobal`, `wm_rexist_reflexive_strength_le`, `wm_dts_ob_pe_strength_le`, `wmRexistReflexiveConsequenceRule`, `wmDtsObPeConsequenceRule`	`Mettapedia/Logic/PLNWorldModelCalculus.lean`, `Mettapedia/PLN/Bridges/WorldModel/PLNWorldModelKripkeConsequence.lean`
WM shell clarifiers (state-level permissive, query-level deterministic) *Tags: [PLNBook-Ch8], [WM], [determinism], [judgment-boundary]*	`WMJudgment.trivial`, `WMJudgment.query_deterministic`, `WMJudgmentCtx.query_deterministic`	`Mettapedia/Logic/PLNWorldModel.lean`
Categorical WM transport layer (institution satisfaction + hyperdoctrine Beck-Chevalley + unified endpoint) *Tags: [PLNBook-Ch8], [WM], [institution], [hyperdoctrine], [Beck-Chevalley], [transport]*	`satisfactionCondition_evidence`, `satisfactionCondition_strength`, `satisfactionCondition_strength_with`, `satisfactionCondition_confidence`, `WMInstitution.sat_condition`, `beckChevalley_transport_world`, `EndpointStatement`, `institution_beckChevalley_endpoint`, `endpointContract_of_hyperdoctrine`, `institution_beckChevalley_endpoint_of`, `ch8_hol_categorical_wrapper`, `ch8_fol_categorical_consequence`	`Mettapedia/Logic/PLNWorldModelInstitution.lean`, `Mettapedia/PLN/Bridges/CategoryTheory/WorldModel/PLNWorldModelHyperdoctrine.lean`, `Mettapedia/PLN/Bridges/CategoryTheory/WorldModelFixture.lean`, `Mettapedia/PLN/Bridges/CategoryTheory/PLNWorldModelCategoricalBridge.lean`, `Mettapedia/PLN/Bridges/CategoryTheory/WorldModel/PLNWorldModelCategoricalRegression.lean`

Two-sort-calculus closed-judgment → WM obligation interface (assumption-parameterized) *Tags: [PLNBook-Ch8], [TwoSortPiSigmaId], [WM], [bridge], [canary]*	`TwoSortClosedTheoryBridge, pureClosedTheoryBridge_to_wm, WMCategoricalEndpointConversion, TwoSortJudgmentWMInterfaceBridge, pureTheoryStep_to_wmStrengthObligation, pureTheoryStep_to_wmStrengthObligation_categorical, pureTheoryStepStar_to_wmStrengthObligation, pureTheoryStepStar_to_wmStrengthObligation_categorical, wmConsequenceRuleOn_of_closed_pureTheoryStep, wmConsequenceRuleOn_of_closed_pureTheoryStep_categorical, wmConsequenceRuleOn_of_closed_pureTheoryStepStar, wmConsequenceRuleOn_of_closed_pureTheoryStepStar_categorical, canary_betaPi_bridge_regression_one_nestedLam_wm, canary_betaPi_bridge_regression_two_nestedLam_wm`
GF syntactic transport → WM obligation adapters (two-sort-interface aligned) + functor boundary witness *Tags: [PLNBook-Ch8], [GF], [WM], [bridge], [category], [expressivity]*	`gfSyntaxHom_to_wmStrengthObligation, gfTreePatternEq_to_wmStrengthObligation, wmConsequenceRuleOn_of_gfSyntaxHom_twoSort, wmConsequenceRuleOn_of_gfTreePatternEq_twoSort, wmConsequenceRuleOn_of_taggedGF_twoSort, syntaxToWMFunctor_not_full, canary_gfSyntaxHom_to_pureStyleObligation, canary_endToEnd_gfRule_to_pureStyleObligation, canary_fregeStrong_to_pureStyleObligation, canary_notFull_constantEvidence`
HOL/FOL implication consequence-transfer wrappers into WM consequence rules *Tags: [PLNBook-Ch8], [HOL], [FOL], [WM-ConsequenceRule], [transfer]*	`wmConsequenceRule_of_pointwise, wmConsequenceRuleOn_of_pointwise, wmConsequenceRuleOn_of_singletonStrengthLE, externalImplication_iff_singletonStrengthLE, multiset_strength_le_of_pointwise_decidable, multiset_strength_le_of_pointwise_fintype, wmConsequenceRuleOn_of_pointwise_categorical, wmConsequenceRuleOn_of_singletonStrengthLE_categorical, pointwiseImpliesOnTheory_of_consequence, pointwiseImpliesOnTheory_iff_singletonStrengthLEOnTheory, consequence_iff_singletonStrengthLEOnTheory, provable_imp_iff_singletonStrengthLEOnTheory, multiset_strength_le_of_consequence, wmConsequenceRuleOn_of_consequence, multiset_strength_le_of_provable_imp, wmConsequenceRuleOn_of_provable_imp, multiset_strength_le_of_consequence_categorical, wmConsequenceRuleOn_of_consequence_categorical, multiset_strength_le_of_provable_imp_categorical, wmConsequenceRuleOn_of_provable_imp_categorical`

Singleton iff-consequence adequacy and multiset lifting *Tags: [PLNBook-Ch8], [Kripke], [Neighborhood], [adequacy], [singleton], [multiset]*	singleton_adequacy_strength_le, pointwiseImplies_iff_singleton_strength_le, multiset_strength_le_of_singleton_strength_le, singleton_consequence_schema_parallel, multiset_lift_schema_parallel	Mettapedia/PLN/Bridges/Logic/WorldModel/PLNWorldModel.lean, Mettapedia/PLN/Bridges/Logic/WorldModel/PLNWorldModelNeighborhood.lean
Kripke proof-theoretic closure (sound/complete implication bridge) *Tags: [PLNBook-Ch8], [Kripke], [Sound], [Complete], [K], [KT], [KD], [K4]*	pointwiseImpliesOn_iff_singleton_strength_le_On, provable_imp_iff_singleton_strength_le_On, provable_imp_iff_singleton_consequence_On, multiset_strength_le_of_provable_imp, multiset_consequence_of_provable_imp, provable_imp_iff_singletonStrengthLEOn_K, provable_imp_iff_singletonStrengthLEOn_KT, provable_imp_iff_singletonStrengthLEOn_KD, provable_imp_iff_singletonStrengthLEOn_K4, multiset_strength_le_of_provable_imp_K, multiset_strength_le_of_provable_imp_KT, multiset_strength_le_of_provable_imp_KD, multiset_strength_le_of_provable_imp_K4, multiset_ob_pe_strength_le_of_provable, multiset_rexist_strength_le_of_provable_KT, multiset_ob_pe_strength_le_of_provable_KD	Mettapedia/PLN/Bridges/Logic/WorldModel/PLNWorldModelKripkeCompleteness.lean
Kripke → neighborhood embedding transport and governance-facing mapped consequence *Tags: [PLNBook-Ch8], [Kripke], [Neighborhood], [embedding], [transport], [governance]*	evidence_mapState_eq, queryStrength_mapState_eq, singletonStrengthLEOnEmbedding_iff, mapState_strength_le_of_provable_imp, mapState_governance_ob_pe_strength_le_of_provable	Mettapedia/PLN/Bridges/Logic/WorldModel/kripke, Mettapedia/PLN/Bridges/Logic/WorldModel/PLNWorldModelKripkeNeighborhoodEmbedding.lean

Canonical Kripke → neighborhood representation and mapped WM consequence corollaries *Tags: [PLNBook-Ch8], [Kripke], [Neighborhood], [canonical], [representation], [governance]*	satisfies_iff_kripke_toNeighborhood, pointed_satisfies_iff, queryStrength_mapState_eq_radModal, singletonStrengthLEOn_canonicalWorldModelKripke, mapState_strength_le_of_provable_imp_canonical, mapState_governance_ob_pe_strength_le_of_provable_canonical, parallel_strength_le_of_provable_imp_canonical, parallel_governance_ob_pe_strength_le_of_provable_canonical, parallel_strength_le_of_provable_imp_KT_EMT_canonical, parallel_strength_le_of_provable_imp_KD_ED_canonical, parallel_governance_rexist_strength_le_KT_EMT_canonical, parallel_governance_ob_pe_strength_le_KD_ED_canonical	Mettapedia/PLN/Bridges/Logic/WorldModel_PLNWorldModelKripkeNeighborhoodCanonical.
Weighted/source-aware Kripke WM specialization and weight-1 transfer theorems *Tags: [PLNBook-Ch8], [Kripke], [weighted], [provenance], [transfer]*	weightedExpansion, weightedCountP_toWeightOne_eq_countP, weightedEvidence_toWeightOne_eq_kripkeEvidence, queryStrength_toWeightOne_eq_kripkeStrength, queryStrength_eq_kripkeExpansion, weighted_strength_le_of_provable_imp, trustedGate, trustedGate_ob_pe_strength_le_of_provable, wmTrustedGateObPeConsequenceRule, toWeightOne_strength_le_of_provable_imp	Mettapedia/PLN/Bridges/Logic/WorldModel_PLNWorldModelKripkeWeighted.
Neighborhood proof-theoretic closure (sound/complete implication bridge) *Tags: [PLNBook-Ch8], [Neighborhood], [Sound], [Complete], [E], [EMN], [EMT], [ED]*	pointwiseImpliesOn_iff_singletonStrengthLEOn, provable_imp_iff_singletonStrengthLEOn, provable_imp_iff_singletonConsequenceOn, multiset_strength_le_of_provable_imp, multiset_consequence_of_provable_imp, provable_imp_iff_singletonStrengthLEOn_E, provable_imp_iff_singletonStrengthLEOn_EMN, provable_imp_iff_singletonStrengthLEOn_EMT, provable_imp_iff_singletonStrengthLEOn_ED, multiset_strength_le_of_provable_imp_E, multiset_strength_le_of_provable_imp_EMN, multiset_strength_le_of_provable_imp_EMT, multiset_strength_le_of_provable_imp_ED	Mettapedia/PLN/Bridges/Logic/WorldModel_PLNWorldModelNeighborhoodCompleteness.

Neighborhood modal/deontic-style consequence refinement and Kripke-parallel endpoints *Tags: [PLNBook-Ch8], [Neighborhood], [WM-ConsequenceRuleOn], [box], [diamond], [governance], [parallel], [K], [C], [Mk], [McK]*	wmConsequenceRule_of_provable_in, Mathematica/PLN/ wmBoxConsequenceRule_of_provable_in, Bridges/Logic/ wmDiaConsequenceRule_of_provable_in, WorldModel/ wm_rexist_reflexive_strength_le_of_EMT, PLNWorldEMTelNeighborhoodConsequence. wm_ob_pe_strength_le_ED, lean wmRexistReflexiveConsequenceRuleNeighborhood, wmDtsObPeConsequenceRuleNeighborhood, wm_k_curried_strength_le_of_provable, wm_c_strength_le_of_provable, wm_mck_strength_le_of_provable, wm_mk_strength_le_of_provable, wm_mck_strength_le_of_axiomMcK, wm_mk_strength_le_of_axiomMk, wmMcKConsequenceRule_of_axiom, wmMkConsequenceRule_of_axiom, wm_k_curried_strength_le_EK, wm_k_curried_strength_le_EMK, wm_c_strength_le_EC, wm_c_strength_le_EMC, governanceModalityToNeighborhood, wm_governance_rexist_strength_le, wm_governance_ob_pe_strength_le, governance_translation_endpoint_block, kripke_neighborhood_rexist_parallel, kripke_neighborhood_ob_pe_parallel
State-indexed governance translation preservation (formula-level recursive map) *Tags: [PLNBook-Ch8], [WMConsequenceRuleOn], [governance], [translation], [preservation], [recursive]*	governanceBoxActionToNeighborhood, Mathematica/PLN/ governanceDiamondActionToNeighborhood, Bridges/Logic/ translateGovernanceFormulaToNeighborhood, WorldModel/ translateGovernanceFormulaToNeighborhood_rexistFormula, PLNWorldModelNeighborhoodConsequence. translateGovernanceFormulaToNeighborhood_obFormula, lean translateGovernanceFormulaToNeighborhood_peFormula, translateGovernanceFormulaToNeighborhood_forbiddenBox, translateGovernanceFormulaToNeighborhood_optionalDiamond, wm_governance_rexist_strength_le_of_translation, wm_governance_ob_pe_strength_le_of_translation, wm_governance_forbidden_shape_of_translation, wm_governance_optional_shape_of_translation, governance_translation_endpoint_block_of_formula, governance_translation_endpoint_block_of_formula_extended

Chapter 8 assumption envelope. For threshold transport in OSLF evidence semantics, disjunction and diamond require the explicit side conditions recorded by the theorem row above: `threshold_or_total` requires total-order comparability of evidence values, and `threshold_dia_total` additionally requires finite branching. Without these conditions, use the boundary theorems

`threshold_or_counterexample` and `threshold_dia_fails`.

Chapter 8 naming scope note (HOL/FOL wrappers). Modules `PLNWorldModelHOLCompleteness.lean` and `PLNWorldModelFOLCompleteness.lean` retain historical `*Completeness` filenames, but the listed Ch. 8 endpoints are consequence-transfer/closure wrappers. The FOL side includes a concrete implication bridge over theory-models; the HOL side is schema-level pending imported concrete proof-system instances.

Chapter 8 categorical regression target. Script-level target for the new WM categorical layer: `scripts/check\_wm\_categorical.sh`.

A.3 Chapter 9 Counterexample-First Theorem Index

Notation used in this table. For a selected premise set S and ground-truth dependency set D: "coverage" means dependencyCoverage$(D, S) = |S \cap D|$ and "card" means $|S|$ (the selected-set cardinality).

Chapter 9 content	Lean theorem(s)	Module
Two-term inclusion-exclusion is not identifiable for union mass *Tags: [PLNBook-Ch9], [multideduction], [identifiability-no-go]*	`same_first_two_terms_different_union_cardinality_ie_sameFirstTwo_differentUnion`	`Mettapedia/Logic/PLNInclusionExclusionIdentifiability.lean`, `Mettapedia/PLN/InferenceControl/Counterexamples/PLNLargeScaleInferenceCounterexamples.lean`
No universal predictor from first-two-term statistics *Tags: [PLNBook-Ch9], [no-universal-estimator]*	`no_universal_union_predictor_from_two_terms_ie_noUniversalPredictor`, `no_single_additive_correction_ie_noSingleAdditiveCorrection`	`Mettapedia/Logic/PLNInclusionExclusionIdentifiability.lean`, `Mettapedia/PLN/InferenceControl/Counterexamples/PLNLargeScaleInferenceCounterexamples.lean`
Multideduction residual decomposition + indexed agreement endpoint *Tags: [PLNBook-Ch9], [multideduction], [positive-core], [residual]*	`unionCard_eq_twoTerm_plus_residual`, `correctedEstimate_agrees_with_indexedUnion_of_residualAssumption`, `repeatedDeductionRevision_agrees_of_residualAssumption`, `no_universal_constant_correction`	`Mettapedia/Logic/PLNInclusionExclusionIdentifiability.lean`

Indexed n-way residual decomposition + indexed agreement family *Tags: [PLNBook-Ch9], [multideduction], [n-way], [positive-core]*	`unionCardN_eq_twoTerm_plus_residual_N`, `correctedEstimateN_agrees_of_residualAssumptionN`, `repeatedDeductionRevisionN_agrees_of_residualAssumptionN`	`Mettapedia/Logic/PLNMultiDeduction.lean`
Trail-free deterministic update can fail to stabilize (2-cycle) *Tags: [PLNBook-Ch9], [protocol-dynamics], [counterexample]*	`orbit_not_eventually_constant`, `trailFree_notEventualConstant`	`Mettapedia/PLN/InferenceControl/ProtocolDynamics/PLNTrailFreeDynamicsCounterexample.lean`, `Mettapedia/PLN/InferenceControl/Counterexamples/PLNLargeScaleInferenceCounterexamples.lean`
Damped SP/SPN positive convergence under eventual fresh evidence *Tags: [PLNBook-Ch9], [protocol-dynamics], [SP/SPN], [convergence]*	`damped_sp_spn_eventually_constant_of_eventual_fresh`, `inconsistency_eventually_zero_of_eventual_fresh`, `dampedOrbit_noFresh_not_eventuallyConstant`	`Mettapedia/PLN/InferenceControl/ProtocolDynamics/PLNTrailFreeDampedConvergence.lean`
Bounded-gap fairness reset bounds for damped SP/SPN *Tags: [PLNBook-Ch9], [protocol-dynamics], [SP/SPN], [fairness], [bound]*	`inconsistency_zero_within_boundedGapFresh`, `inconsistency_zero_within_eventually_boundedGapFresh`	`Mettapedia/PLN/InferenceControl/ProtocolDynamics/PLNTrailFreeDampedConvergence.lean`
Pooling axioms are insufficient to characterize additive fusion *Tags: [PLNBook-Ch9], [pooling], [non-uniqueness]*	`maxPoolingOperator_ne_fusion`, `poolingAxioms_notUnique`	`Mettapedia/PLN/InferenceControl/PremiseSelection/ExternalBayesianity.lean`, `Mettapedia/PLN/InferenceControl/Counterexamples/PLNLargeScaleInferenceCounterexamples.lean`

Corrected uniqueness: external-Bayes + total-additivity recovers additive pool *Tags: [PLNBook-Ch9], [pooling], [corrected-positive]*	`poolE_eq_hplus_of_externalBayes_totalAdd`, `poolingUnique_of_externalBayes_totalAdd`	`Mettapedia/PLN/InferenceControl/PremiseSelection/ExternalBayesianity.lean`, `Mettapedia/PLN/InferenceControl/Counterexamples/PLNLargeScaleInferenceCounterexamples.lean`
Coverage/cardinality surrogate is insufficient for risk-sensitive selection *Tags: [PLNBook-Ch9], [premise-selection], [surrogate-no-go]*	`sameCoverageAndCard_differentRisk`, `no_universal_risk_predictor_from_coverage_and_card`, `coverageCard_not_sufficient_for_selection`, `noUniversalRiskPredictorFromCoverageCard`	`Mettapedia/PLN/InferenceControl/PremiseSelectionNegative.lean`, `Mettapedia/PLN/InferenceControl/Counterexamples/PLNLargeScaleInferenceCounterexamples.lean`
Finite BRGI object + policy-sensitive optimum characterization *Tags: [PLNBook-Ch9], [BRGI], [positive-core], [optimization], [policy]*	`mem_feasibleSets_iff`, `exists_optimal_feasible_set`, `exists_optimal_feasible_set_iff`, `optimal_value_eq_min_of_feasible`, `brgi_toy_optimal_value`	`Mettapedia/PLN/InferenceControl/PremiseSelectionPolicy_characterization.lean`
Graph-level BRGI object + optimality-preserving refinement wrappers *Tags: [PLNBook-Ch9], [BRGI], [graph-object], [refinement], [policy]*	`exists_optimal_feasible_set_brg`, `exists_optimal_feasible_set_iff_brg`, `brg_optimal_value_eq_min_feasible`, `brg_graph_toy_optimal_value`	`Mettapedia/PLN/InferenceControl/PremiseSelectionBRGI.lean`
Class-packaged BN d-separation discharge (non-counterexample positive path) *Tags: [PLNBook-Ch9], [BN-bridge], [assumption-packaging]*	`allDiscreteCPTLocalMarkovOfFintype`, `wmqueryeq_of_dsep_discharge`, `chain_screeningOff_wmquery_eq_of_dsep_allLocalMarkov`, `fork_screeningOff_wmquery_eq_of_dsep_allLocalMarkov`, `collider_screeningOff_wmquery_eq_of_dsep_allLocalMarkov`	`Mettapedia/PLN/Bridges/ProbabilityTheory/BayesNetBridgeReadiness_allLocalMarkov.lean`, `Mettapedia/PLN/Bridges/ProbabilityTheory/BayesNetDischarge.lean`, `Mettapedia/PLN/Bridges/ProbabilityTheory/BayesNet/PLNBNLocalMarkovPackages.lean`

One-call selector → rewrite → threshold composition *Tags: [PLNBook-Ch9], [selector], [WM-OSLF-threshold], [composed-endpoint]*	`selector_rewrite_threshold_end_to_end_of_interval`, `bool_selector_rewrite_threshold_end_to_end_fixture`	`Mettapedia/PLN/InferenceControl/PLNCanonicalAPI.lean`, `Mettapedia/PLN/InferenceControl/SelectorRewriteThreshold/PLNSelectorRewriteThresholdExamples.lean`
Chain/fork/collider single-call concrete WM→OSLF→threshold endpoints *Tags: [PLNBook-Ch9], [BN-bridge], [single-call], [concrete]*	`xi_deduction_threshold_concrete_true_fixture`, `xi_sourceRule_threshold_concrete_true_fixture`, `xi_sinkRule_threshold_concrete_true_fixture`	`Mettapedia/PLN/FirstOrder/PLNXiDerivedBNRules.lean`
Independent-cause union canary ("at least one head in n biased flips") *Tags: [PLNBook-Ch9], [independent-causes], [noisy-or], [executable-canary]*	`noisyOrMulti_replicate`, `atLeastOneHeadProb_eq_noisyOr_replicate`, `atLeastOneHeadProb_coin4_p06`, `atLeastOneHeadProb_coin4_p01`	`Mettapedia/Logic/PLNNoisyOr.lean`
Stochastic experiment layer: Blackwell factorization, weighted source policies, trusted-gate transfer *Tags: [PLNBook-Ch9], [experiment], [Blackwell], [weighted-policy], [trusted-gate]*	`expectedUtility_eq_of_blackwellFactor`, `optimalValue_mono_of_blackwellFactor`, `optimalValueWeighted_mono_of_blackwellFactor`, `optimalValuePolicy_mono_of_blackwellFactor`, `optimalValue_trustedGate_mono_of_blackwellFactor`, `fixture_expectedUtility_eq_of_blackwellFactor`, `fixture_optimalValue_mono`, `fixture_trustedPolicy_optimalValue_mono`	`Mettapedia/PLN/World/BlackwellFactor.lean`, `Mettapedia/PLN/World/Model.lean`, `Mettapedia/PLN/World/ModelExperiment.lean`, `Mettapedia/PLN/World/ModelExperimentStochastic.lean`, `Mettapedia/PLN/World/ModelExperimentStochasticRegression.lean`

Chapter 9 MeTTa/PeTTa Usage Snippet (theorem-name anchors only).

```
xi_deduction_threshold_concrete_true_fixture
xi_sourceRule_threshold_concrete_true_fixture
xi_sinkRule_threshold_concrete_true_fixture
selector_rewrite_threshold_end_to_end_of_interval
bool_selector_rewrite_threshold_end_to_end_fixture
brgi_toy_optimal_value
repeatedDeductionRevision_agrees_of_residualAssumption
repeatedDeductionRevisionN_agrees_of_residualAssumptionN
damped_sp_spn_eventually_constant_of_eventual_fresh
inconsistency_zero_within_bound_of_boundedGapFresh
brg_graph_toy_optimal_value
```

```
noisyOrMulti_replicate
atLeastOneHeadProb_eq_noisyOr_replicate
expectedUtility_eq_of_blackwellFactor
optimalValue_mono_of_blackwellFactor
optimalValuePolicy_mono_of_blackwellFactor
fixture_trustedPolicy_optimalValue_mono
```

Chapter 9 Session Checklist (summary-template).

- Counterexample-first coverage checked: inclusion-exclusion, pooling, trail-free dynamics, coverage/card risk.
- Positive BN bridge coverage checked: chain/fork/collider d-separation to WM/OSLF/threshold endpoints.
- Multideduction positive core checked: explicit residual decomposition + assumption-indexed agreement endpoint.
- Multideduction n-way extension checked: indexed-family decomposition + indexed agreement family.
- SP/SPN positive dynamics checked: eventual-fresh damping convergence + inconsistency-to-zero endpoint.
- SP/SPN fairness bounds checked: bounded-gap reset theorem with explicit ‘B+1‘ window.
- BRGI object coverage checked: feasible-set gate, objective bounds, and finite optimality endpoint.
- BRGI graph refinement checked: graph-object wrappers preserve policy-sensitive optimum characterization.
- Independent-cause canary checked: noisy-or replicate and finite concrete points.
- Stochastic experiment layer checked: Blackwell utility monotonicity plus weighted/trusted-source policy transfer.
- Selector→rewrite→threshold endpoint checked (canonical + worked fixture).
- Note: operational runtimes are not implemented in Lean here; this section tracks theorem-level semantic/bridge coverage.

A.4 Chapter 10 Rule-to-Theorem Index

Chapter 10 content	Lean theorem(s)	Module
SatisfyingSet bridge: higher-order evaluation to first-order membership *Tags: [PLNBook-Ch10], [HOI-FOI-bridge], [SatisfyingSet]*	`member_eq_evaluation`	`Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`
Extensional implication as subset relation *Tags: [PLNBook-Ch10], [Implication], [Subset]*	`extensional_implication_reduces_to_subset`	`Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`
Implication-conditioned inheritance value reduction *Tags: [PLNBook-Ch10], [Implication], [Inheritance], [value-level]*	`inheritance_reflects_implication`, `implication_reduces_to_inheritance_value`, `implication_reduces_to_pointwiseImplicationPred`, `forAll_himp_to_inheritance_ratio_of_bridge`	`Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`
Unconditional chapter implication-value equality is false (counterexample) *Tags: [PLNBook-Ch10], [Implication], [counterexample], [boundary]*	`forAll_himp_not_equal_inheritance_ratio_unconditional`	`Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`
Unconditional chapter equivalence-value equality is false (counterexample) *Tags: [PLNBook-Ch10], [Equivalence], [counterexample], [boundary]*	`forAll_equivalence_not_equal_inheritance_ratio_unconditional`	`Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`
Exact self-division fixed-point characterization (four p-bit corners) *Tags: [PLNBook-Ch10], [value-level], [fixed-point], [corners]*	`ennreal_self_div_component_fixed_iff`, `self_div_fixed_iff_pbit_corners`	`Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`

Piecewise iff closures for chapter implication/equivalence ratio endpoints *Tags: [PLNBook-Ch10], [Implication], [Equivalence], [iff], [piecewise]*	`forAll_himp_ratio_iff_piecewise`, `forAll_equivalence_ratio_iff_piecewise`	`Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`
Side-condition packaging and endpoint ergonomics (single bundled assumptions) *Tags: [PLNBook-Ch10], [API], [side-conditions], [ergonomics]*	`HimpRatioSideConditions`, `EquivalenceRatioSideConditions`, `himp_sideConditions_of_true`, `equivalence_sideConditions_of_true`	`Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`, `forAll_himp_to_inheritance_ratio_of_sideConditions`, `forAll_equivalence_to_similarity_ratio_of_sideConditions`
Inference-rule wrappers tied to ratio-endpoint soundness *Tags: [PLNBook-Ch10], [inference-rule], [soundness], [API]*	`HimpReductionRule`, `HimpReductionRule.conclusion`, `HimpReductionRule.sound`, `EquivalenceReductionRule`, `EquivalenceReductionRule.conclusion`, `EquivalenceReductionRule.sound`	`Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`
Equivalence-conditioned similarity value reduction (first proved bridge) *Tags: [PLNBook-Ch10], [Equivalence], [Similarity], [value-level]*	`HOEquivalent`, `pointwiseEquivalencePred`, `equivalence_reduces_to_similarity_value`, `forAll_equivalence_to_similarity_ratio_of_bridge`, `similarity_symmetric`	`Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`
Constructive corner fixtures and worked endpoint instantiation *Tags: [PLNBook-Ch10], [fixtures], [corners], [worked-example]*	`corner_source_unit_true`, `corner_source_unit_bot`, `implication_ratio_fixture_unit_both`, `equivalence_ratio_fixture_unit_both`	`Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`

HOJ/HOS/Hyp/Context micro-semantics and soundness layer *Tags: [PLNBook-Ch10], [HOJ], [HOS], [Hyp], [Context], [soundness]*	`HigherOrderStatement`, `HigherOrderJudgment`, `hosSemantics`, `hojSemantics`, `hojAlignment`, `hoj_sound_of_alignment`, `HypAtom`, `andExtSet`, `ContextMacroPayload`, `expandContextMacro`, `context_macro_semantics_sound`, `TruthCleanPred`, `hosSemantics_andExt_of_srcTruthClean`, `context_macro_sound_of_hos_srcTruthClean`, `context_macro_sound_of_hoj_alignment_srcTruthClean`	`Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`
Concrete Context/Hyp fixtures (math/juggling + mismatch boundary) *Tags: [PLNBook-Ch10], [Context], [Hyp], [fixtures], [boundary]*	`BenContextObj`, `ctx_payload_math`, `ctx_payload_juggling`, `ctx_payload_mismatch`, `context_hyp_fixture_math_sound`, `context_hyp_fixture_juggling_sound`, `context_hyp_fixture_mismatch_not_sound`	`Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`
Subset-conditioned inheritance structural collapse *Tags: [PLNBook-Ch10], [Subset], [Inheritance], [structural]*	`subset_implies_strong_inheritance`	`Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean`

A.5 Chapter 11 Rule-to-Theorem Index

Chapter 11 rule/content	**Lean theorem(s)**	**Module**
Quantifier exchange (De Morgan) *Tags: [PLNBook-Ch11], [Classical-DeMorgan]*	`canary_exists_deMorgan`, `canary_quantifier_exchange`	`Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierCanary.lean`

Existential generalization (extensional) *Tags: [PLNBook-Ch11], [Classical-FO]*	`canary_existential_generalization`	`Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierCanary.lean`
Universal specification (extensional) *Tags: [PLNBook-Ch11], [Classical-FO]*	`canary_universal_specification`	`Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierCanary.lean`
Fuzzy existential generalization (score positivity) *Tags: [PLNBook-Ch11], [Zadeh1983]*	`fuzzyExistsScore_pos_of_witness`, `fuzzyExistsScore_pos_of_strength`	`Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzyQuantifierSemantics.lean`, `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzyITVBridge.lean`
Fuzzy universal specification (‘PCL = 1‘) *Tags: [PLNBook-Ch11], [Zadeh1983]*	`nearOne_of_fuzzyForAll_one`, `nearOne_itvStrength_of_fuzzyForAll_one`	`Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzyQuantifierSemantics.lean`, `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzyITVBridge.lean`
Rule-4 non-equivalence witness (ITV path) *Tags: [PLNBook-Ch11], [ITV-counterexample]*	`canary_existentialConjunction_not_equivalent_itvPath`	`Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierWorkedExamples.lean`

ITV coordinate transfer (lower/upper ⇒ strength) *Tags: [PLNBook-Ch11], [Walley-IDM], [QFM-bridge]*	`fuzzyIntervalHolds_strength_le_of_le`, `canary_itv_strength_interval_lower_upper`	`Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzyITVBridge.lean`, `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierWorkedExamples.lean`
ITV rule-family bridge (existential generalization, universal specification, exchange) *Tags: [PLNBook-Ch11], [ITV-bridge], [QFM-bridge]*	`fuzzyThereExistsHolds_itv_rule_family_core`, `canary_itv_rule_family_bridge`	`Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzyITVBridge.lean`, `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierWorkedExamples.lean`
Finite MOST/FEW fixtures *Tags: [Zadeh1983], [Yager1984]*	`canary_most_fraction_three_quarters`, `canary_few_fraction_one_quarter`, `canary_most_vs_few_split_on_three_quarters`	`Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierWorkedExamples.lean`
Zadeh-style quantified syllogism family (‘MOST ∘ MOST‘, ‘FEW ∘ MOST‘) *Tags: [Zadeh1983], [Yager1984]*	`zadeh_syllogism_most_most`, `zadeh_syllogism_few_most`, `canary_zadeh_most_most_figure`, `canary_zadeh_few_most_figure`	`Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzySyllogismCanary.lean`

QFM abstraction + generic composition transport *Tags: [DiazHermida2018-QFM], [QFM-Axiomatic]*	QFMCompose, qfmMul, qfmMin, qfmLukasiewicz, qfmProbSum, qfm_compose_interval_bundle, qfm_compose_interval_of_fuzzyInterval, qfmMul_interval_of_fuzzyIntervals, qfmMin_interval_of_fuzzyIntervals, qfmLukasiewicz_interval_of_fuzzyIntervals, qfmProbSum_interval_of_fuzzyIntervals, qfm_syllogism_interval, qfm_syllogism_most_most, qfm_syllogism_few_most	Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzyQuantifierSemantics.lean, Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzySyllogismCanary.lean
QFM-oriented canaries/schemas (monotonicity, conservativity-style signature invariance) *Tags: [DiazHermida2018-QFM], [Tserkovny2020]*	nearZero_iff_nearOne_one_sub, nearZeroFraction_eq_nearOneFraction_of_one_sub, fuzzyThereExistsHolds_iff_nearZero_complement, fuzzyExistsScore_mono_of_pointwise, fuzzyForAllHolds_mono_of_pointwise, nearOneFraction_eq_of_signatureEq, fuzzyIntervalHolds_iff_of_signatureEq, qfm_composedScore_mono_of_pointwise, qfm_composedScore_eq_of_signatureEq, qfm_composedInterval_iff_of_signatureEq, canary_qfm_monotonicity_mean, canary_qfm_forall_monotonicity_most, canary_qfm_conservativity_same_nearOne_signature	Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzyQuantifierSemantics.lean, Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierEq.lean, Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzySyllogismCanary.lean
Additional finite quantifier fixtures (‘ALMOST ALL‘, ‘ABOUT HALF‘)	*Tags: [Zadeh1983], [Glöckner2006]* canary_almostAll_fraction_threeQuarters, canary_almostAll_holds_on_threeQuarters, canary_aboutHalf_fraction_oneHalf, canary_aboutHalf_positive_negative_split	Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierWorkedExamples.lean
Literature-calibrated verbal probability fixtures (‘PROBABLY‘, ‘MAYBE/POSSIBLY‘)	*Tags: [MostellerYoutz1989], [IPCC-uncertainty-guidance]* probablyParams, canary_probably_positive_negative_split, canary_probably_holds_on_threeQuarters, canary_probably_rejects_oneHalf, maybeParams, canary_maybe_positive_negative_split, canary_maybe_holds_on_oneHalf	Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierWorkedExamples.lean

Additional finite quantifier fixtures (‘ALMOST NONE‘, ‘MANY-BUT-NOT-MOST‘)	*Tags: [Zadeh1983], [Yager1984], [DiazHermida2018-QFM]*	`Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzySyllogismCanary.lean`
QFM selector-bundle API endpoints *Tags: [QFM-Axiomatic], [PLNBook-Ch11]*	`qfm_selector_bundle_mostMinusFew`, `qfm_selector_bundle_few_mostMinusFew`, `qfm_selector_bundle_mostFirstOrder`, `qfm_selector_bundle_few_mostFirstOrder`, `qfmMin_syllogism_most_most`, `qfmMin_syllogism_few_most`	`Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzySyllogismCanary.lean`
Non-multiplicative QFM instance theorem families *Tags: [Lukasiewicz-tnorm], [Probabilistic-sum], [QFM-instance-comparison]*	`qfmLukasiewicz_syllogism_most_most`, `qfmLukasiewicz_syllogism_few_most`, `qfmProbSum_syllogism_most_most`, `qfmProbSum_syllogism_few_most`, `qfm_instance_comparison_fuzzySyllogism`, `qfm_instance_comparison_few_most`	`Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzySyllogismCanary.lean`

Regression entrypoint	`lake build Mettapedia.PLN.RuleFamilies.FirstOrder.Quantifiers.QuantifierRegression`, `lake build Mettapedia.PLN.RuleFamilies.FirstOrder.Quantifiers.QuantifierRegression Mettapedia.PLN.RuleFamilies.FirstOrder.Quantifiers.FuzzySyllogismRegressionFin`, `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierRegression.lean`, `Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzySyllogismRegression.lean`	`.github/workflows/lean\_action\_ci.yml`

A.6 FO/HO Steelman Survey (PLN-Side)

PLN book statement class	**Strongest steelmanned formal form**	**Lean theorem(s)**
Chapter 5 FO extensional implication rules	Explicit probabilistic derivation under measurable-space assumptions (not heuristic-only formulas)	`pln_deduction_from_total_probability`, `pln_deduction_from_total_probability_ctx`, `pln_deduction_decomposition`, `pln_deduction_uniform_simplifies`
Chapter 10 HO implication/equivalence value equalities	Unconditional equalities are false; strongest valid endpoints are side-conditioned bridges plus piecewise iff closures	`forAll_himp_not_equal_inheritance_ratio_unconditio`, `forAll_equivalence_not_equal_similarity_ratio_unco`, `forAll_himp_to_inheritance_ratio_of_sideConditions`, `forAll_equivalence_to_similarity_ratio_of_sideCond`, `forAll_himp_ratio_iff_piecewise`, `forAll_equivalence_ratio_iff_piecewise`

Chapter 10 HOJ/HOS/Hyp/Context layer	Context soundness exposed with explicit source-truth-clean side conditions and concrete positive/negative fixtures	`context_macro_sound_of_hoj_alignment_srcTruthClean`, `context_hyp_fixture_math_sound`, `context_hyp_fixture_juggling_sound`, `context_hyp_fixture_mismatch_not_sound`
Chapter 11 quantifier layer	Finite-model FO + fuzzy/QFM endpoint families with monotonicity and conservativity-style transport	`fuzzyExistsScore_mono_of_pointwise`, `fuzzyForAllHolds_mono_of_pointwise`, `qfm_composedScore_mono_of_pointwise`, `qfm_composedScore_eq_of_signatureEq`, `qfm_composedInterval_iff_of_signatureEq`
HOL/FOL consequence transfer into WM/PLN transport (categorical contract)	Besides concrete HOL/FOL categorical fixtures, FOL now has a full implication bridge `T |- (phi -> psi) <-> singleton consequence on T-models`	`multiset_strength_le_of_pointwise_categorical`, `provable_imp_iff_singletonStrengthLEOnTheory`, `ch8_hol_categorical_wrapper_bool_fixture`, `ch8_fol_categorical_consequence_singleton_fixture`

Scope note (Chapter 11). Current Chapter 11 quantifier theorems are intentionally finite-model (`[Fintype U]`) results. The strongest present formal claims are finite FO/fuzzy-QFM statements above; infinitary FO/HOL quantifier extensions remain separate future layers.

A.7 Chapter 12 Rule-to-Theorem Index

Chapter 12 content	Lean theorem(s)	Module
Information-theoretic bridge to universal-mixture conditionals *Tags: [PLNBook-Ch12], [Solomonoff], [log-ratio]*	`intensionalFromConditionalTemplate_eq_log2_ratio`, `intensionalFromXiSemimeasure_eq_log2_ratio`, `intensionalFromXiGeom_eq_log2_ratio`	`Mettapedia/Logic/InformationTheoreticRelevanceSolomonoffBridge.`
Typed WM ASSOC/PAT channel grounding *Tags: [PLNBook-Ch12], [WM-typed], [OSLF-bridge]*	`AssocScoreCorrespondence`, `PATScoreCorrespondence`, `mixedPolicyAssoc_of_assocScoreCorrespondence`, `mixedPolicyAssocPat_of_assocPatScoreCorrespondence`	`Mettapedia/Logic/PLNIntensionalWorldModel.`

Canonical semantic packaging for concrete mixed policies *Tags: [PLNBook-Ch12], [ASSOC-core], [semantic-packaging]*	`AssocSemanticModel`, `AssocPatSemanticModel`, `mixedPolicyAssoc_of_assocSemanticModel`, `mixedPolicyAssocPat_of_assocPatSemanticModel`, `mixedRewriteRule_assocSemantic_of_model`, `mixedRewriteRule_assocPatSemantic_of_model`	`Mettapedia/Logic/PLNIntensionalWorldModel.lean`
Threshold-truth transport (semantic linked strong endpoint) *Tags: [PLNBook-Ch12], [ITV], [selector-transport]*	`intensional_mixed_assoc_atom_of_interval_of_solomonoff_semantic_end_to_end_quantale_selector_rewrite_query_threshold_finalBundle_end_to_end_quantale_selector_rewrite_query_threshold_finalBundle_end_to_end_quantale_selector_rewrite_query_threshold_finalBundle`	`Mettapedia/Logic/PLNIntensionalWorldModel.lean`
Linked-strong selector-specialized theorem wrappers *Tags: [PLNBook-Ch12], [wrapper], [bayesNormal/bayesExact/walley]*	`intensional_mixed_assoc_atom_bayesNormal_of_solomonoff_semantic, intensional_mixed_assoc_atom_bayesExact_of_solomonoff_semantic, intensional_mixed_assoc_atom_walley_of_solomonoff_semantic`	`Mettapedia/Logic/PLNIntensionalWorldModel.lean`
Concrete ASSOC+PAT semantic fixture endpoints *Tags: [PLNBook-Ch12], [ASSOC+PAT], [concrete-end-to-end]*	`end_to_end_assocPat_bayesNormal_lower, end_to_end_assocPat_bayesExact_lower, end_to_end_assocPat_walley_lower`	`Mettapedia/Logic/PLNIntensionalAssocPatClosure.lean`
Canonicality no-go witness for binary collapse *Tags: [PLNBook-Ch12], [ASSOC+PAT], [no-go], [canonicality]*	`no_binary_mixed_policy_factorization`	`Mettapedia/Logic/PLNIntensionalAssocPatClosure.lean`
Executable canaries (positive + non-equivalence) *Tags: [PLNBook-Ch12], [canary], [non-equivalence]*	`canary_mixed_extensional_projection, canary_mixed_assoc_projection_nonequivalent, canary_end_to_end_bayesNormal_lower_threshold, canary_end_to_end_bayesExact_lower_threshold, canary_end_to_end_walley_lower_threshold`	`Mettapedia/Logic/PLNIntensionalCanary.lean`
Regression entrypoint	`Mettapedia/Logic/PLNIntensionalRegression.lean`, `lake build Mettapedia.PLN.ConceptGeometry.AssocPat.PLNIntensionalRegression`	`Mettapedia/Logic/README.md`

A.8 Chapter 13 Rule-to-Theorem Index

Chapter 13 content	Lean theorem(s)	Module
Selector defaults + checklist linkage *Tags: [PLNBook-Ch13], [selector-spec], [checklist]*	selectorSpec_defaults_mathematical, selectorSpec_default_priorNeutral, selectorSpec_zeroLocalGap_iff_pick	Mettapedia/PLN/InferenceControl/PremiseSelection/SelectorSpec.lean
Ranking transfer from classical baselines *Tags: [PLNBook-Ch13], [NB], [kNN], [ranking-transfer]*	pln_inherits_nb_ranking, pln_inherits_knn_ranking, pln_knn_ranking_eq	Mettapedia/PLN/InferenceControl/PremiseSelection/Optimality.lean
Robustness under bounded perturbations *Tags: [PLNBook-Ch13], [ranking-stability], [margin]*	pairwise_lt_stable_of_margin, bayesRanking_stable_of_margin, bayesRanking_stable_of_supportSelectionMargin	Mettapedia/PLN/InferenceControl/PremiseSelection/RankingStability.lean
Coverage/submodularity objective family *Tags: [PLNBook-Ch13], [submodularity], [greedy-coverage]*	dependencyCoverage_mono, dependencyCoverage_diminishingReturns, greedyChain_one_minus_exp_bound	Mettapedia/PLN/InferenceControl/PremiseSelection/Coverage.lean
Composed Chapter-13 core endpoint (selector + stability + coverage) *Tags: [PLNBook-Ch13], [composed-core], [end-to-end]*	selector_default_ranking_iff, stable_pooled_of_twoStage, inferenceControl_end_to_end	Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlCore.lean
Executable greedy algorithm endpoint *Tags: [PLNBook-Ch13], [algorithmic], [end-to-end]*	inferenceControl_end_to_end_algorithmic, greedySelect_chain_of_le_card, greedySelect_one_minus_exp_bound	Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlAlgorithms.lean
Operator-style forward/backward chainer interfaces *Tags: [PLNBook-Ch13], [chainer], [MeTTa/PeTTa-style]*	forwardStep, forwardSearch, backwardStep, backwardSearch, boundedSearch, forwardSearch_chain_of_le_card, boundedSearch_eq_forwardSearch_of_card_le_topK	Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlChainer.lean

PeTTa-style propositional connection-tableau ATP model *Tags: [PLNBook-Ch13], [ATP], [connection-tableau], [proof-trace-checker], [executable-DFS], [PeTTa-aligned]*	`replay_sound`, `replay_complete`, `checkProof_true_iff_root_mem_and_connDerivable`, `checkProof_sound`, `exists_checkProof_true_of_root_mem_and_connDerivable`, `connProveDFS_sound`, `connProveDFS_complete_of_trace`, `connProveDFS_witness_complete`, `op_check_proof_sound`	`Mettapedia/Logic/LP/PropositionalConnectionChainer.lean`
PeTTa fixture-parity map (`test_leancop_*` naming) *Tags: [PLNBook-Ch13], [fixture-map], [parity], [ic_unsat], [ic_branching]*	`ic_unsat_checkProof_true`, `ic_sat_connProve_none`, `ic_unsat_connProve_has_proof`, `ic_branching_two_distinct_roots_proofs`, `ic_branching_negp_connProve_has_proof`, `ic_branching_negq_connProve_has_proof`	`Mettapedia/Logic/LP/PropositionalConnectionChainer.lean`
First-order proof-trace checker soundness (`proof_fo` shape) *Tags: [PLNBook-Ch13], [ATP], [FO], [proof-trace-checker], [substitution]*	`traceDerivableFO_of_replayFO_proof`, `replayFO_sound`	`Mettapedia/Logic/LP/FirstOrderConnectionTrace.lean`
FO fixture map + branch-complete root-membership (PeTTa `fo_*` naming) *Tags: [PLNBook-Ch13], [ATP], [FO], [fixture-map], [collectProofsAll]*	`fo_unsat_ground`, `fo_unsat_unify`, `fo_sat_singleton`, `proof_mem_collectProofsAllFO_of_root_mem_trace_mem`, `fo_unsat_ground_proof_mem_collectProofsAllFO`, `fo_unsat_unify_proof_mem_collectProofsAllFO`	`Mettapedia/Logic/LP/FirstOrderConnectionTrace.lean`
Operator-contract parity rows for bundle/topK outputs (`op_leancop_ch13_bundle`) *Tags: [PLNBook-Ch13], [ATP], [operator-contract], [topK], [bundle]*	`rootProofBundle_proofs_eq_collectProofs`, `rootProofBundle_topK_eq_topKCollectProofs`, `branchProofBundle_proofs_eq_collectProofsAll`, `branchProofBundle_topK_eq_topKCollectProofsAll`, `mem_branchProofBundle_topK_implies_mem_branchProofBundle_proofs`, `topKCollectProofsAllFO_complete`, `mem_topKCollectProofsAllFO_implies_mem_collectProofsAllFO`	`Mettapedia/Logic/LP/FirstOrderConnectionTrace.lean`

Worked finite fixtures (‘topK=1‘ and nontrivial ‘topK=2‘) *Tags: [PLNBook-Ch13], [worked-example], [finite-fixture]*	`bool_end_to_end_algorithm`, `bool_coverage_exact_one`, `fin3_end_to_end_algorithm`, `fin3_coverage_exact_two`	`Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlExamples.lean`
Executable canaries (positive + negative) *Tags: [PLNBook-Ch13], [canary], [non-equivalence]*	`canary_pln_inherits_nb_revision`, `canary_pairwise_stable_of_margin`, `canary_ranking_flip_with_premise_gain`, `canary_dependencyCoverage_insufficient_for_control`, `canary_selector_default_gain`	`Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlCanary.lean`
Regression entrypoint	`Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlRegression.lean`, `lake build Mettapedia.PLN.InferenceControl.PremiseSelection.PLNInferenceControlRegression`	`Mettapedia/Logic/README.md`

A.9 Module-Level Coverage Index (Legacy + Recent)

To make explicit that coverage is not limited to recent additions, this index lists the principal *legacy and current* modules in Mettapedia that track PLN-book content.

Ch. 3–4 (experiential semantics, indefinite truth values). `Mettapedia/Logic/EvidenceClass.lean`, `Mettapedia/Logic/EvidenceQuantale.lean`, `Mettapedia/Logic/PLNIndefiniteTruth.lean`, `Mettapedia/Logic/PLNWorldModelITV.lean`, `Mettapedia/Logic/PLNWorldModelITVHypercube.lean`.

Ch. 5 (FO extensional inference: rules and formulas). `Mettapedia/Logic/PLNDeduction.lean`, `Mettapedia/PLN/RuleFamilies/FirstOrder/PLNDerivation.lean`, `Mettapedia/Logic/PLNInferenceRules.lean`, `Mettapedia/Logic/PLNConjunction.lean`, `Mettapedia/Logic/PLNRevision.lean`, `Mettapedia/Logic/PLNImplicantConjunction.lean`, `Mettapedia/Logic/PeTTaLibPLNTruthFunctions.lean`, `Mettapedia/Logic/WMPLNJustifiedTruthFunctions.lean`, `Mettapedia/PLN/Comparisons/PeTTa/PeTTaLibPLNFormalAnalysis.lean`, `Mettapedia/Logic/PLNBayesNetFastRules.lean`, `Mettapedia/PLN/Bridges/ProbabilityTheory/BayesNet/PLNBNCompilation.lean`, `Mettapedia/PLN/RuleFamilies/FirstOrder/PLNXiDerivedBNRules.lean`.

Ch. 6–7 (ITV/distributional truth-value layers). `Mettapedia/Logic/PLNDistributional.lean`, `Mettapedia/PLN/Bridges/ProbabilityTheory/EvidenceBeta`.

lean, Mettapedia/PLN/Bridges/ProbabilityTheory/EvidenceDirichlet.lean, Mettapedia/Logic/HigherOrder/PLNKyburgReduction.lean, Mettapedia/PLN/Bridges/ProbabilityTheory/EvidenceNormalGamma.lean, Mettapedia/PLN/Bridges/Languages/PLNWMOSLFBridgeITV.lean, Mettapedia/PLN/Core/PLNCanonicalAPI.lean.

Ch. 8 (error magnification / diagnostics). Mettapedia/Logic/PLNBugAnalysis.lean, Mettapedia/Logic/Comparison/ErrorCharacterization.lean, Mettapedia/PLN/Core/SoundnessCompleteness.lean, Mettapedia/Logic/PLNSoundnessCounterexample.lean, Mettapedia/Logic/PLNSoundnessDiagnosis.lean.

Ch. 9 and Ch. 13 (large-scale inference + inference control). Mettapedia/PLN/InferenceControl/PremiseSelection/KNN.lean, Mettapedia/PLN/InferenceControl/PremiseSelection/KNN_PLNBridge.lean, Mettapedia/PLN/InferenceControl/PremiseSelection/Optimality.lean, Mettapedia/PLN/InferenceControl/PremiseSelection/Coverage.lean, Mettapedia/PLN/InferenceControl/PremiseSelection/RankingStability.lean, Mettapedia/PLN/InferenceControl/PremiseSelection/SelectorSpec.lean, Mettapedia/PLN/InferenceControl/PremiseSelection/BestPLNDraft.lean, Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlCore.lean, Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlCanary.lean, Mettapedia/PLN/InferenceControl/PremiseSelection/PLNInferenceControlRegression.lean, lake build Mettapedia.PLN.InferenceControl.PremiseSelection.PLNInferenceControlRegressi

Ch. 10–11 (higher-order, quantifiers, crisp/fuzzy operators). Mettapedia/Logic/HigherOrder.lean, Mettapedia/Logic/HigherOrder/Basic.lean, Mettapedia/Logic/HigherOrder/HigherOrderReduction.lean, Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers.lean, Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzyQuantifierSemantics.lean, Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/FuzzyITVBridge.lean, Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierCanary.lean, Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierWorkedExamples.lean, Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/QuantifierRegression.lean, Mettapedia/PLN/RuleFamilies/FirstOrder/Quantifiers/Soundness.lean, Mettapedia/Logic/PLNCrispEvidence.lean, Mettapedia/Logic/PLNNoisyOr.lean, Mettapedia/Logic/PLNNegation.lean, Mettapedia/Logic/PLNDisjunction.lean, lake build Mettapedia.PLN.RuleFamilies.FirstOrder.Quantifiers.QuantifierRegression.

Ch. 12 (intensional inference). Mettapedia/Logic/IntensionalInheritance.lean, Mettapedia/Logic/IntensionalInheritanceSolomonoffBridge.lean, Mettapedia/Logic/PLNIntensionalWorldModel.lean, Mettapedia/PLN/Core/PLNCanonicalAPI.lean, Mettapedia/Logic/PLNIntensionalCanary.lean, Mettapedia/Logic/PLNIntensionalRegression.lean, lake build Mettapedia.PLN.ConceptGeometry.AssocPat.PLNIntensionalRegression.

Ch. 14 (temporal/causal; event calculus; OSLF/GSLT grounding). `Mettapedia/PLN/RuleFamilies/Temporal/PLNTemporal.lean`, `Mettapedia/Logic/TemporalQuantale.lean`, `Mettapedia/PLN/RuleFamilies/Temporal/PLNTemporalCausalInference.lean`, `Mettapedia/PLN/RuleFamilies/Temporal/PLNProbabilisticEventCalculus.lean`, `Mettapedia/OSLF/Framework/VertexTemporalRewriteRules.lean`, `Mettapedia/OSLF/Framework/HypercubeTemporalGSLTFunctor.lean`, `Mettapedia/OSLF/Framework/QuantaleCoherence.lean`.

Appendix A (PLN/NARS comparison). `Mettapedia/PLN/Comparisons/NARS/PLNNARSRuleCorrespondence.lean`, `Mettapedia/PLN/Comparisons/NARS/NARSEvidenceBridge.lean`, `Mettapedia/PLN/Comparisons/NARS/NARSPLNGaloisConnection.lean`.

This module-level index is intended as the explicit "not just the new files" map.

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